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Fact-Checking Election-Campaign Misinformation: Impacts on Noncommitted Voters' Feelings and Behavior

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Abstract

Making misleading statements may benefit a politician, for example during an election campaign. However, there are potentially also negative consequences; political misinformation can taint democratic debate, voters may be misled into forming false beliefs, and being fact-checked may damage a politician's reputation. Previous research has found that correcting misleading statements made by established politicians reduces topical misperceptions, but hardly affects voter feelings and support. Here, we examined the impact of political misinformation and fact-checking when politicians are unfamiliar. Participants (N = 406) were engaged in a simulated election campaign set in an unfamiliar country, featuring statements from fictional candidates. Participants indicated their feelings toward the candidates, cast a vote, and rated their belief in the fact-checked statements. Misleading statements that were not corrected positively affected feelings towards and voting for (right-leaning) politicians. Corrective fact-checks had large effects, reducing belief in misinformation, and fact-checked candidates were viewed much less favorably and attracted far fewer votes. This demonstrates that in the absence of strong pre-existing attitudes, corrective fact-checks can negatively impact misinformation-spreading politicians who are not (yet) well-known.

Keywords: misinformation, fact-checking, political attitudes, belief change, voting behavior

Fact-Checking Election-Campaign Misinformation: Impacts on Noncommitted Voters' Feelings and Behavior

Misinformation—defined as any information that is misleading or proven false—can have widespread negative consequences (e.g., Ecker et al., 2022; Schmid et al., 2023; Simonov et al., 2022; Swire-Thompson & Lazer, 2022; Tay et al., 2024). Political misinformation can be particularly damaging. It has the potential to reduce public trust and decrease satisfaction with democracy (Nisbet et al., 2021; Ross et al., 2022), and contribute to real-world violence, such as the storming of the U.S. Capitol on January 6, 2021 (Lee et al., 2022; Riley, 2022). Political misinformation is also caustic because it can potentially influence voters (Gunther et al., 2019; Zimmermann & Kohring, 2020; although see Leyva & Beckett, 2020), undermining their ability to make well-informed and evidence-based decisions. This is important because open and authentic, good-faith public debate is a crucial component of a functioning democracy, and such debate can be undermined by misinformation (Gilens, 2001; Kuklinski et al., 2000; Lewandowsky et al., 2017; Tenove, 2020). In line with these concerns, recent years have seen increased research attention on political misinformation and its impact on political processes (e.g., Hameleers & Minihold, 2020; Lewandowsky et al., 2020, 2023, 2024; Lippman et al., 2016). Here, we focus on the impact of misleading statements made by politicians and how people respond to attempts to correct these statements, both directly (e.g., by updating their beliefs) and indirectly (e.g., by changing their attitudes toward the politicians).

One common approach to countering misinformation is fact-checking, in which an article or post suspected to contain misinformation is retroactively labelled as true or false (Barrera et al., 2020; Ecker et al., 2020; Nyhan & Reifler, 2016). This label is sometimes accompanied by a brief message explaining why a claim is true or false, ideally citing a trusted source (Vraga &

Bode, 2017). Although fact-checking is generally effective at reducing misinformation belief, it often fails to completely undo the impact of misinformation exposure. This tendency for people to continue to rely on corrected misinformation to some extent when drawing inferences or making decisions is known as the continued influence effect (Ecker et al., 2022; Johnson & Seifert, 1994).

While there is some evidence that political worldview (i.e., a person's compendium of prior beliefs, ideological commitments, and partisan identity) can influence the efficacy of a corrective fact-check (Ecker & Ang, 2019; Nyhan & Reifler, 2010; also see Gaines et al., 2007; Taber & Lodge, 2006; Weeks, 2015), subsequent research has failed to replicate these results (Ecker et al., 2021; Wood & Porter, 2019). Additionally, several papers have shown that fact-checking claims from politicians leads both supporters and non-supporters of those politicians to update their beliefs (Aird et al., 2018; Nyhan et al., 2020; Prike et al., 2023; Swire, Berinsky, et al., 2017; Swire-Thompson et al., 2020). Therefore, despite some ongoing debate, there is an emerging consensus that fact-checks generally reduce belief in false claims regardless of worldview congruence (Berinsky, 2017; Prike & Ecker, 2023; Swire-Thompson et al., 2020).

Although the direct impact of corrections on beliefs is reasonably well-established, the indirect effects of political misinformation and corrections on political attitudes and decisions (e.g., support for candidates, voting intentions, and voting decisions) are much less clear. Within a U.S. context, studies have found that although both supporters and non-supporters of former President Trump updated their beliefs in response to fact-checks of his statements, there was very limited impact on voting intentions and feelings of warmth toward Trump (Swire, Berinsky, et al., 2017; Swire-Thompson et al., 2020). This was replicated across the political aisle, with the similar results emerging when statements of former presidential primary candidate Bernie

Sanders were used (Swire-Thompson et al., 2020). However, this pattern of findings seems to be at least somewhat country-specific, with a study by Aird et al. (2018) showing that when statements of political leaders were fact-checked in an Australian context, there was a modest but significant impact on voting intentions and feelings of warmth—although only when the ratio of false to true statements was 4:1, not when it was 4:4. A recent study by Prike et al. (2023) replicated this finding with U.K. participants and politicians.

One potential explanation for the discrepancies may be that the U.S. political environment is more polarized (Jenke, 2023), which may have resulted in participants having comparatively stronger pre-existing attitudes toward candidates, limiting the impact of fact-checking on voters' feelings or voting intentions. Participants may have also differed in their expectations around politicians' truthfulness, as trust in government is lower in the U.S. than either Australia or the U.K., according to the OECD (2021). Thus, U.S. voters may expect their representatives to lie more often, and therefore voter attitudes may be less impacted by fact-checking (Swire-Thompson et al., 2020).

Present Study

The present study aimed to further investigate how misinformation and subsequent fact-checks impact voter perceptions, feelings, behaviors, and beliefs. The study expanded upon previous work in several key areas. First, most prior studies have used statements made by real political figures (Aird et al., 2018; Nyhan & Reifler, 2016; Swire, Berinsky, et al., 2017). Although this has the advantage of increased realism and relevance, participants are likely to have considerable pre-existing knowledge of and attitudes toward the politicians, which limits our ability to experimentally examine the potential impact of misinformation and corrections in the absence of such strong priors. To this end, we created two fictional centrist political

candidates (one slightly left-leaning, one slightly right-leaning) engaged in an election campaign that was set in Malta. Malta was chosen because it is a Western democracy and has English as an official language, but Maltese politics were likely to be unfamiliar to U.S. participants. We recruited politically moderate participants, who were presented with a series of social-media statements by and about the candidates in an impression-formation paradigm (Ecker & Rodricks, 2020). This allowed participants to form opinions and attitudes about the politicians throughout the experiment, as one might do when learning about a new politician who is gaining in prominence or upon moving to a new country or location and engaging with an unfamiliar political environment.

Second, we aimed to make presentation of materials more realistic by using a social-media simulator (Butler et al., 2023), as though participants were experiencing an election campaign. With voters commonly encountering political misinformation on social media—and news organizations, political organizations, and politicians regularly disseminating information via social-media platforms (Lewandowsky et al., 2020; Nunziato, 2020)—the simulator provided a realistic environment in which to study the effects of political misinformation and corrections. After the simulation, participants were required to cast a vote for one of the two candidates; this not only served to further enhance realism and motivate participants to form an impression of the candidates, but also provided a behavioral measure, thus enabling us to go beyond the self-report measures commonly used in misinformation research, which often do not allow strong conclusions regarding behaviors (McEachan et al., 2011).

The statements that candidates made included generic, non-verifiable statements typical of those made during election campaigns, as well as statements deemed to be true or false. True and false statements were designed to positively influence perceptions of candidates'

competence and morality, two central dimensions of person impressions (Abele & Wojciszke, 2007; Bennett et al., 2019; Lausten & Bor, 2017; Mickelberg et al., 2022; Pfau & Diedrich, 1993; Redlawsk, 2006). Competence refers to an individual's perceived intelligence, efficiency, and capability, whereas morality refers to an individual's perceived honesty, loyalty, and cooperativeness. During the social-media simulation, the true and false statements were displayed to participants (in all conditions but the control condition); in the fact-check conditions, fact-checks were displayed alongside these statements as comments made by a fact-checking group. Fact-checks could affirm true statements or correct false statements. In line with previous studies, the ratio of statements identified as false versus true in the fact-check conditions was 4:1 (Aird et al., 2018; Swire-Thompson et al., 2020). We measured warmth felt toward candidates and perceptions of candidates' competence and morality. Participants were then asked to vote for one of the candidates, before rating their belief in the presented statements.

We made several key predictions. First, we predicted that misinformation would enhance positive feelings (warmth) and perceptions (competence and morality) toward the relevant candidate relative to the opposing candidate (H1A), and that fact-checks would reduce those feelings and perceptions (H1B). Second, it was predicted that misinformation would increase the number of votes for a particular candidate relative to the opposing candidate (H2A), while exposure to fact-checks would reduce the number of votes (H2B). For these hypotheses, the key comparisons were how relative feelings, perceptions, and voting behavior (i.e., the difference between the left- and right-leaning candidates) differed between the control condition and the four experimental conditions. Third, it was hypothesized that presenting claims would slightly increase belief in those claims (an illusory truth effect; Fazio et al., 2015; Pennycook et al., 2017; H3A), and that fact-checks would lead to reduced belief in false claims (H3B) and increased

belief in true claims (H3C). Because corrections often do not entirely undo the impact of misinformation exposure (Ecker et al., 2022; Johnson & Seifert, 1994), a continued influence effect was anticipated, but this was not a key hypothesis.

Method

This study used a 2 × 2 plus control between-subjects design. Participants received a set of social-media posts featuring statements made by or about two fictional politicians running in an upcoming general election set in Malta. A cover story was provided asking participants to imagine they had recently moved to Malta, that they were eligible to vote in an upcoming election, and that they had turned to social media for more information on the candidates to decide who to vote for (it was also mentioned that election participation in Malta is very high). There were 16 generic statements (eight per candidate), which were used to give a baseline impression of each candidate, and to position the candidates as slightly left- or right-leaning; these were presented in all conditions, and were the only statements shown in the control condition. The experimental conditions additionally included eight false statements, all associated with one of the candidates, designed to enhance that candidate's perceived competence and morality or reduce the perceived competence and morality of their opponent, as well as four true statements (two per candidate). All statements came from either the candidates themselves, or from a fictional Maltese newspaper (the "Maltese Observer"). In the fact-check conditions, fact-checks of all true and false statements were provided by a fictional fact-checking organization ("Fact Checks Malta"), which was introduced as nonpartisan and objective. Thus, one design factor ("misinformation candidate"; left vs. right) was defined by whether the misinformation favored the left-leaning or right-leaning candidate, and the other factor was the

presence of fact-checks ("fact-check"; no vs. yes). For clarity, Table 1 provides an overview of the statements displayed across conditions.

Participants

An a-priori power analysis (using G*Power 3; Faul et al., 2007) suggested a minimum sample size of 310 to detect small effects ($f = 0.20$, with $\alpha = 0.05$ and $1 - \beta = .80$). To ensure sufficient power to conduct follow-up planned contrasts even after potential exclusions, we recruited 455 participants. Participants were U.S.-based Prolific users (www.prolific.com) who met the platform's prescreening criterion of identifying as politically moderate. Participants were randomly assigned to one of the five conditions, with the constraint of approximately equal group sizes. Prior to analyses, participants were removed who claimed to have recognized either of the candidates ($n = 8$) or made multiple attempts at completing the survey ($n = 1$). Participants with strong left or right political orientation (as measured within the survey; for details see below) were also removed ($n = 40$). Final sample size was thus $N = 406$ ($M_{age} = 38.72$ years, $SD_{age} = 13.58$; 218 males, 187 females, 1 non-binary person). The experiment took approximately 15 minutes and participants were reimbursed US$2.50.

Table 1

Overview of Statements Provided in Each Condition

Candidate	Control	Misinformation candidate: left; fact-check: no	Misinformation candidate: right; fact-check: no	Misinformation candidate: left; fact-check: yes	Misinformation candidate: right; fact-check: yes
Left-leaning	8 generic statements	8 generic, 8 false, 2 true statements	8 generic, 2 true statements	8 generic, 8 false/fact-checked, 2 true/fact-checked statements	8 generic, 2 true/fact-checked statements
Right-leaning	8 generic statements	8 generic, 2 true statements	8 generic, 8 false, 2 true statements	8 generic, 2 true/fact-checked statements	8 generic, 8 false/fact-checked, 2 true/fact-checked statements
Total statements shown	16	28	28	28	28

Materials

Politician Profiles

Profiles of two fictional politicians were provided (see Supplementary Materials), indicating their names ("Michael Vella"; "Roberto Camilleri") and which of two fictional parties they belonged to ("Democratic Alliance"; "United Democratic Party"). The profiles also included headshots; we used images of two actual European politicians relatively unknown in the U.S., as determined by a Google Trends analysis (Robert Abela, Prime Minister of Malta; Borut Pahor, former President of Slovenia). Assignment of candidate names, party names, and headshots to left- and right-leaning candidates was fully counterbalanced across participants. Counterbalancing also controlled for the order in which candidates were introduced (which matched the order in which participants assessed the candidates at test). This resulted in 80 different survey versions (16 per experimental condition).

Statements

Statements included in the study were chosen based on a pilot study conducted on an initial pool of 42 statements (20 generic, 16 deemed false, 6 deemed true). A separate group of participants ($N = 52$) rated each statement's perceived plausibility and political valence (on -2 to +2 Likert scales), and the extent to which it would shift their perceptions of the candidates' relative morality and competence (on -5 to +5 sliding scales; see Table 2 and Supplementary Materials). Initially, statements with mean plausibility ratings $M < 0.5$ were removed. Then, the four true and eight false statements associated with the largest competence and morality shifts were selected, with the constraint that the overall mean political valence of the true and false statements, respectively, remained close to zero. Sixteen generic statements that fit best with the chosen true and false statements were selected, and

then organized into two sets: one slightly left-leaning (mean political-valence score M = -0.26; range -1.71 – 0.60) and one slightly right-leaning (M = 0.25; range -1.02 – 1.38).

Statements were presented as social-media posts using a simulator designed to study the processing of information on social media (Butler et al., 2023). Posts were presented on individual pages and accompanied by a profile picture and the handle of the individual or organization making the statement (i.e., “Michael Vella”, “Roberto Camilleri”, or “Maltese Observer”). Participants were able to interact with posts by “liking” or “disliking” the post, commenting on it, liking others’ comments, or skipping it. These interactions were included for the sake of realism and were not analyzed. Example posts are shown in Figure 1.

Measures

Candidate Measures

Three dimensions were measured to assess participants’ impressions: perceived competence, perceived morality, and warmth felt toward candidates. Perceived competence and morality were measured using an 11-point sliding scale (e.g., “Considering Roberto Camilleri as a candidate for prime minister, how would you rate his competence [morality]?” – “0, Very incompetent” to “10, Very competent” [“0, Very immoral” to “10, Very moral”]). Feelings of warmth were measured using an 11-point thermometer sliding scale (e.g., “Please use the thermometer below to indicate how warmly you feel about Roberto Camilleri” – “0, Very cold” to “10, Very warm”). Sliders were initially anchored at the scale midpoint.

Voting Behavior

Participants were asked to vote for one of the candidates by selecting one of two horizontally-aligned radio buttons labelled with the candidate and party names. Prior to voting, participants were shown the profiles of the two candidates again to remind them of each candidate’s name, face, and party.

Table 2

True, False, and Generic Statements Used in the Experiment

False Statements	
Statement	**Fact-check**
"As treasurer, Candidate X loaned millions of Euros to foreign interest groups – we still haven't seen that money back! He lacks financial competence."	FALSE. Candidate X did not hand out loans. He was the administrator of a fund that invested public money in multinational infrastructure projects, as part of Malta's foreign aid program.
"A competent leader must face challenges such as climate change head-on. I have always supported a clean-energy future for our nation. My voting history reflects this."	FALSE. Public voting records show that Candidate X has voted against multiple measures designed to support clean energy, while voting in favor of some measures deemed to have negative climate impacts.
"The Maltese people have indicated consistently, in all polls, that they believe Candidate X is most competent to lead the nation."	FALSE. In polls that asked voters about competence, results have been inconsistent. Polling results about who will win the election have likewise been mixed.
"I take our relations with our neighbors very seriously. Since entering parliament, I have traveled to each and every Mediterranean Leadership Summit."	FALSE. Travel records show that Candidate X has only travelled to two out of the five summits held since he entered parliament.
"Candidate X was once a member of a communist student organization. He doesn't hold the right values to lead the nation."	FALSE. Candidate X once gave a speech at an event co-organized by the Communist Students' League but was never affiliated with them.
"We need a leader who understands how small business works from the ground up – Candidate X has owned and operated his own business for 20 years."	FALSE. While Candidate X has been involved in the family business, he has neither owned nor operated any business.
"I know what it's like to have very little. I understand the needs of working families, so I can govern for everyone."	FALSE. Candidate X grew up in a middle-class household, and received a large inheritance as a teenager.
"During my time in the private sector, I received multiple awards for leadership excellence. I have the experience needed to lead this country."	FALSE. Candidate X was nominated or self-nominated for some leadership awards but did not win any.
True Statements	
Statement	**Fact-check**
"During my time as minister, crime rates dropped to the lowest they've been in over 10 years."	TRUE. Crime rates dropped by around 2% during Candidate X's term as minister, making them marginally lower than in any period in the preceding 12 years.
"Corruption is a big problem in our country – we need to solve it. I've voted in favor of	TRUE. According to parliamentary records, Candidate X has a strong record of voting in favor of anti-corruption laws.

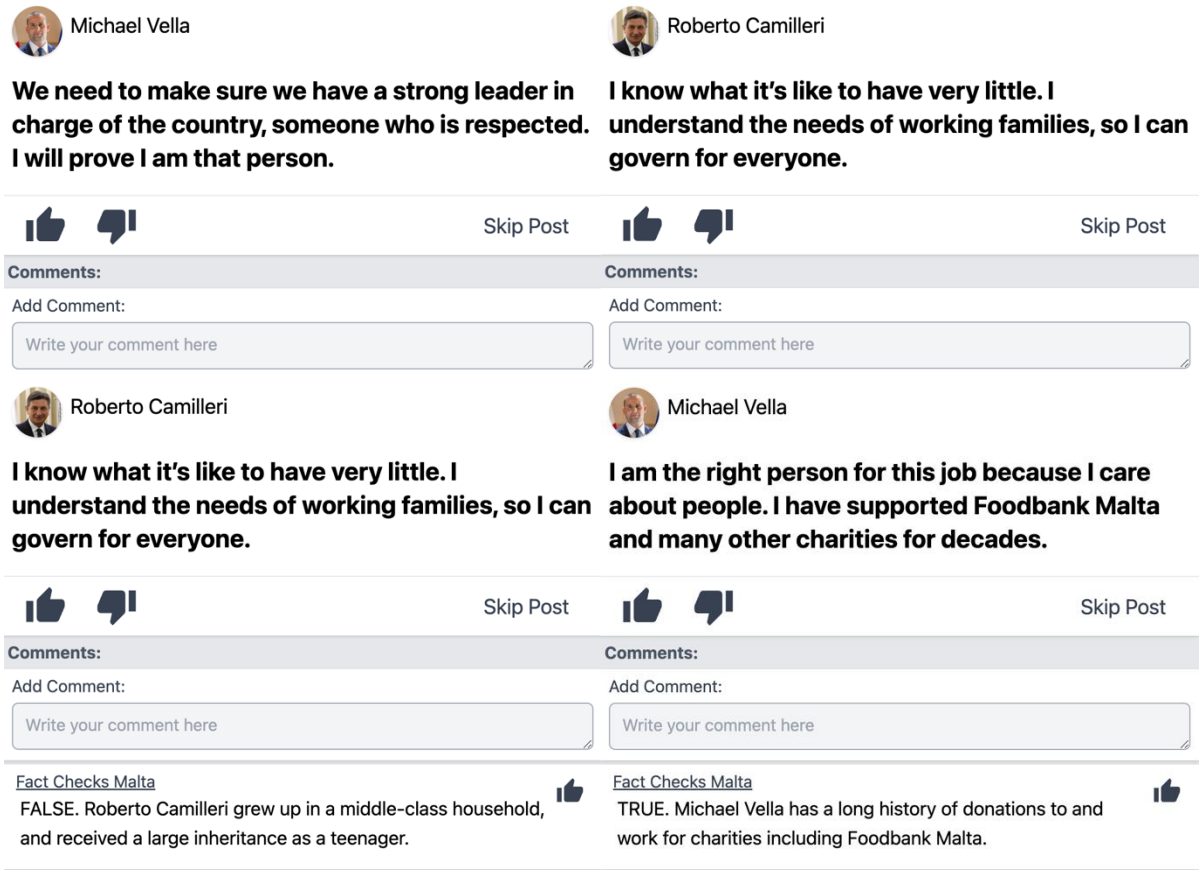
anti-corruption laws every time they've been on the table."	
"Under my stewardship as Minister for Employment, unemployment rates in residents over the age of 50 dropped significantly."	TRUE. While Candidate X was Minister for Employment, unemployment rates in older workers dropped by around 2.1%.
"I am the right person for this job because I care about people. I have supported Foodbank Malta and many other charities for decades."	TRUE. Candidate X has a long history of donations to and work for charities including Foodbank Malta.

Generic Statements

Left-leaning candidate	**Right-leaning candidate**
"Equality is important for our future. We will do everything in our power to make sure no-one is left behind. Our country's success should benefit everyone."	"We need to be tougher on crime and border protection. We will invest in our police and our armed forces – they have been neglected for too long!"
"Our children need a healthy environment to live in. We need to focus on sustainability!"	"To maintain the prosperity of our country we need to look at things that have worked for us in the past. Let us be guided by our shared values and traditions!"
"Coming into office my government will improve the public transport system. No more late buses and trains."	"Our nation is, and always has been, the most important thing. I will put Malta first!"
"We will break down barriers to development, and will build a brighter future together!"	"Growing up I always looked up to the leaders of our country. It will be an honor for me to follow in their footsteps."
"My father taught me growing up that honesty is always an important quality. As prime minister, I intend to be honest with our people every day."	"Our government will focus on jobs and growth! If I am elected I will bring unemployment rates down!"
"I feel proud to have served as a minister. I would like nothing more than to bring my expertise to the position of Prime Minister."	"Come out to the rally tomorrow night, I'm always excited to meet my supporters!"
"We're going to win this election and make the country a better place!"	"To thrive in the future, we need to get each and every child in Malta the best education possible. If I am elected, I will invest in the public education system!"
"We need to make sure we have a strong leader in charge of the country, someone who is respected. I will prove I am that person."	"We need affordable health care. We will strengthen the healthcare system so it works better for everyone!"

Figure 1

Example Posts



Note. The figure shows (from top-left to bottom-right): a generic post, a post deemed false with no fact-check, the same false post with fact-check, and a post deemed true with fact-check.

Statement Belief

Belief in the true and false statements was measured via questionnaire, using 11-point Likert scales ("0, Entirely untruthful" to "10, Entirely truthful"). Although participants in the control group were not exposed to these statements, their beliefs were also measured to provide a baseline.

Political Ideology

To confirm that participants were actually politically moderate (as indicated on Prolific), political ideology was measured using six of the seven items of Mehrabian's (1996)

conservatism-liberalism scale.[1] Participants responded to each statement on a seven-point Likert scale (e.g., “Socialism has many advantages over capitalism” – “0, Strongly disagree” to “6, Strongly agree”).

Procedure

Testing was administered online using Qualtrics (Qualtrics, Provo, UT, USA). All participants received an ethics-approved information sheet and provided informed consent. Participants answered a brief demographic questionnaire, followed by completing the social-media simulation, with statements presented in random order. They then completed the candidate-impression, voting behavior, and statement-belief measures, in that order, followed by the conservatism-liberalism scale, before being fully debriefed. Testing occurred in August 2022.

Results

To simplify the analyses, and because we were primarily interested in how misinformation and fact-checks impacted impressions of candidates relative to each other, we calculated difference scores for competence, morality, and warmth. This was done by subtracting competence, morality, and warmth scores for the right-leaning candidate from the equivalent score for the left-leaning candidate. Difference-score calculations were conducted separately for each participant and each candidate-impression measure.

To ensure robustness, all analyses were repeated without excluding any participants and, separately, with the inclusion of age, gender, and political ideology as predictors (see https://osf.io/wfbck/?view_only=959ded2da87441d19f7498a20dfebc04). Only the competence planned contrast between the control and misinformation-left conditions differed

[1] An item referring to communism was excluded because it was deemed irrelevant.

across analysis specifications. This planned contrast was significant when no participants were excluded (p = .027) but not in the main analyses (p = .050). All other results were consistent across analysis specifications.

Competence

Competence scores (see Figure 2) were analyzed in a one-way between-subjects ANOVA, with experimental condition as the factor. The condition factor subsumed the design factors fact-check (no vs. yes) and misinformation candidate (i.e., the candidate releasing or potentially benefiting from the misinformation). There was a significant main effect of condition, $F(4, 401) = 51.31$, $p < .001$ $\eta_p^2 = .34$, 95% CI [.26, .40].

For the competence aspect of the first hypothesis, we predicted that relative to the control condition, misinformation in the absence of fact-checks would lead to greater perceptions of competence for the misinformation candidate relative to their opponent ($H1A_{Comp}$), but that fact-checks would reduce competence perceptions ($H1B_{Comp}$) relative to control. To test this, four planned contrasts were performed in R using the *emmeans* package, with Holm-Bonferroni corrections for multiple comparisons applied (Lenth, 2023). The first contrast tested whether competence difference scores differed between the control and misinformation-left conditions. This contrast was non-significant, $t(401) = 1.97$, p =.050, d = 0.31, 95% CI [-0.001, 0.62]; thus, relative to control, the presence of misinformation supporting the left-leaning candidate did not significantly change their perceived competence relative to the right-leaning candidate. The second contrast tested whether competence difference scores differed significantly between the control and misinformation-right conditions. This contrast was significant, $t(401) = 3.67$, p = .001, d = 0.58, 95% CI [0.27, 0.89]; thus, relative to control, when there was misinformation benefiting the right-leaning candidate, they were perceived as more competent than the left-leaning candidate. This partially supported hypothesis $H1A_{Comp}$.

The third contrast tested whether competence difference scores differed between the control and fact-check left conditions. This was significant, $t(401) = 5.79$, $p < .001$, $d = 0.90$, 95% CI [0.59, 1.21]; thus, relative to control, when misinformation favoring the left-leaning candidate was presented but fact-checked, they were perceived as less competent than the right-leaning candidate. The fourth contrast tested this difference between the control and fact-check right conditions. This contrast was also significant, $t(401) = 7.24$, $p < .001$, $d = 1.14$, 95% CI [0.82, 1.46]; relative to control, when misinformation favoring the right-leaning candidate was presented but fact-checked, they were perceived as less competent than the left-leaning candidate. $H1B_{Comp}$ was thus supported. It is also noted that fact-checks reduced perceptions of competence to substantially below control levels.

In sum, misinformation slightly improved relative perceptions of competence, although only significantly so for right-leaning candidates. Fact-checks significantly and substantially reduced these perceptions.

Morality

The analogous ANOVA on morality ratings (see Figure 3) yielded the same pattern: a main effect of condition, $F(4, 401) = 66.58$, $p < .001$, $\eta_p^2 = .40$, 95% CI [.33, .46].

A contrast analysis was again performed to test the morality aspect of the first hypothesis, namely that misinformation would enhance perceptions of relative morality ($H1A_{Mor}$) and that fact-checks would reduce those perceptions ($H1B_{Mor}$). The first contrast tested whether morality difference scores differed between the control and misinformation-left conditions. The contrast was significant, $t(401) = 2.43$, $p = .015$, $d = 0.38$, 95% CI [0.07, 0.69]; misinformation favoring the left-leaning candidate thus led to them being perceived as significantly more moral than the right-leaning candidate, relative to control. The second contrast tested the same effect with the misinformation-right and control conditions. The contrast was also significant, $t(401) = 2.80$, $p = .011$, , $d = 0.44$, 95% CI [0.13, 0.75]; thus, the

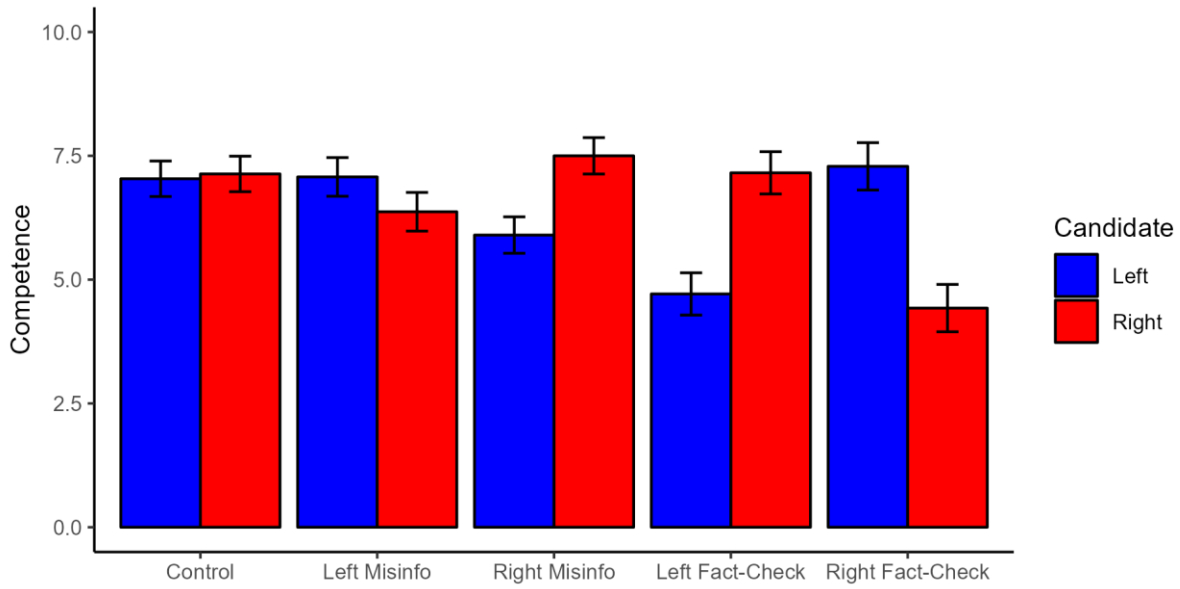
perceived-morality difference between the right-leaning and left-leaning candidates was significantly greater, relative to control, when there was misinformation favoring the right-leaning candidate. $H1A_{Mor}$ was thus supported.

The third contrast tested the difference between the control and left fact-check conditions. The contrast was significant, $t(401) = 7.14$, $p < .001$, $d = 1.11$, 95% CI [0.80, 1.43]; when misinformation favoring the left-leaning candidate was presented and fact-checked, the left-leaning candidate was perceived as significantly less moral than the right-leaning candidate, relative to control. The fourth contrast tested the same effect with the right fact-check condition. The contrast was significant, $t(401) = 8.32$, $p < .001$, , $d = 1.31$, 95% CI [0.99, 1.63]; thus, compared to control, the right-leaning candidate was perceived as significantly less moral than the left-leaning candidate when there was fact-checked misinformation favoring the right-leaning candidate. $H1B_{Mor}$ was thus supported. Again, fact-checks reduced perceptions of morality to well below control levels.

In sum, misinformation slightly improved perceptions of relative morality. Fact-checks significantly and substantially reduced these perceptions.

Figure 2

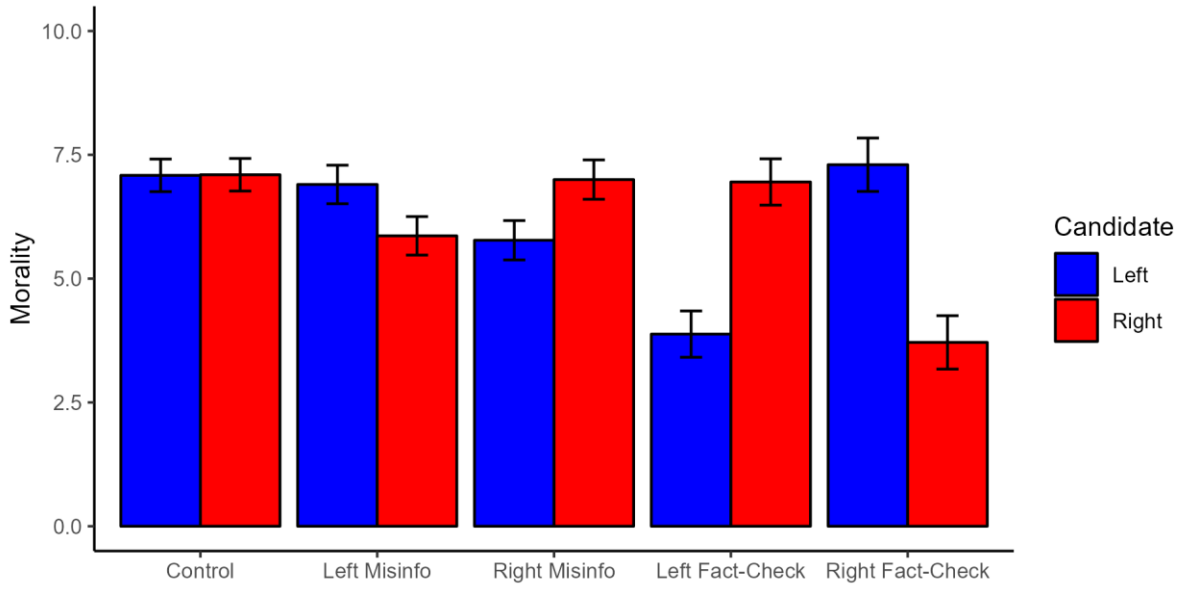
Mean Competence Scores for Left- and Right-Leaning Candidates Across Conditions



Note. Left/Right Misinfo, condition with misinformation favoring the left/right-leaning candidate, respectively; Left/Right Fact-Check, condition with misinformation favoring the left/right-leaning candidate but fact-checked. Errors bars represent 95% confidence intervals.

Figure 3

Mean Morality Scores for Left- and Right-Leaning Candidates Across Conditions



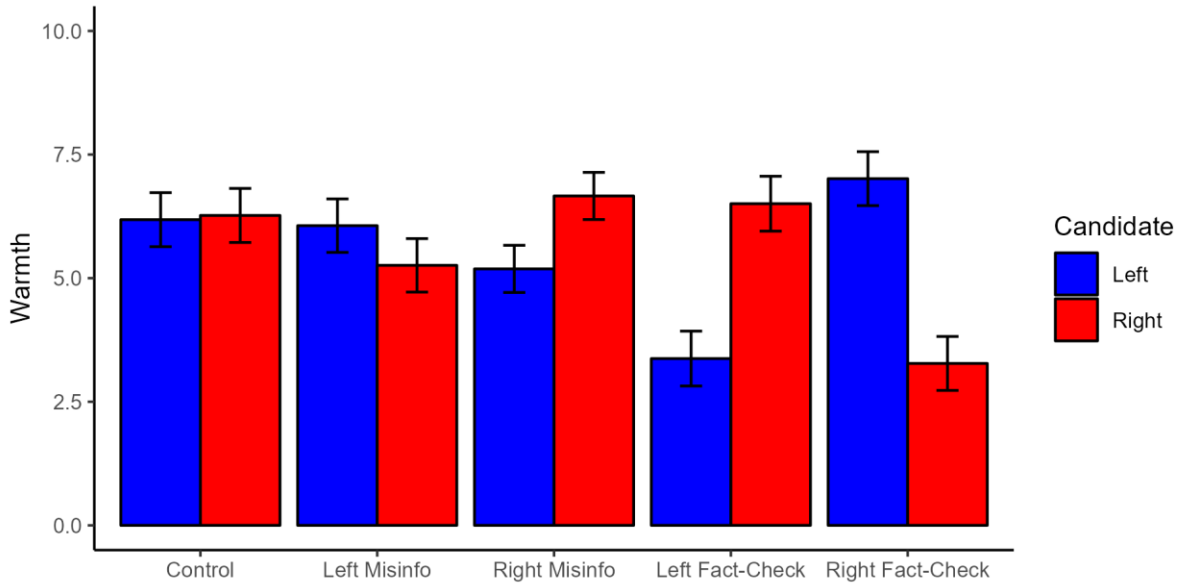
Note. Left/Right Misinfo, condition with misinformation favoring the left/right-leaning candidate, respectively; Left/Right Fact-Check, condition with misinformation favoring the left/right-leaning candidate but fact-checked. Errors bars represent 95% confidence intervals.

Warmth

The analogous ANOVA on warmth ratings (see Figure 4) yielded the same outcome, a main effect of condition, $F(4, 401) = 46.08$, $p < .001$, $\eta_p^2 = .31$, 95% CI [.24, .38].

Figure 4

Mean Warmth Scores for Left- and Right-Leaning Candidates Across Conditions



Note. Left/Right Misinfo, condition with misinformation favoring the left/right-leaning candidate, respectively; Left/Right Fact-Check, condition with misinformation favoring the left/right-leaning candidate but fact-checked. Errors bars represent 95% confidence intervals.

A contrast analysis was again conducted to test the warmth aspect of the first hypothesis, namely that misinformation would enhance relative feelings of warmth ($H1A_{Warm}$) and that fact-checks would reduce those feelings ($H1B_{Warm}$). The first contrast tested the difference in relative warmth between the control and misinformation-left conditions. The contrast was non-significant, $t(401) = 1.66$, $p = .098$, , $d = 0.26$, 95% CI [-0.05, 0.57]; thus, the difference in warmth between the left- and right-leaning candidates was not significantly greater than control when there was misinformation favoring the left-leaning candidate. The second contrast tested the same effect with the misinformation-right condition. The contrast was significant, $t(401) = 2.59$, $p = .020$, , $d = 0.41$, 95% CI [0.10, 0.72]; relative to control, when there was misinformation favoring the right-leaning

candidate, participants felt more warmly about them than the left-leaning candidate. $H1A_{Warm}$ was thus partially supported.

The third contrast tested the difference in warmth between the control and fact-check left conditions. The contrast was significant, $t(401) = 5.72$, $p < .001$, , $d = 0.89$, 95% CI [0.58, 1.20]; thus, relative to control, participants felt less warmly about the left-leaning than the right-leaning candidate when there was misinformation favoring the left-leaning candidate that was fact-checked. The fourth contrast tested the same effect with the fact-check right condition. The contrast was significant, $t(401) = 7.11$, $p < .001$, , $d = 1.12$, 95% CI [0.80, 1.44]; thus, relative to control, participants felt less warmly about the right-leaning than the left-leaning candidate when there was misinformation favoring the right-leaning candidate that was fact-checked. $H1A_{Warm}$ was thus supported. Again, it is noted that fact-checks reduced warmth to well below control levels.

In sum, misinformation slightly improved warmth felt toward candidates relative to opponents, although only significantly so for right-leaning candidates. Fact-checks significantly and substantially reduced these feelings.

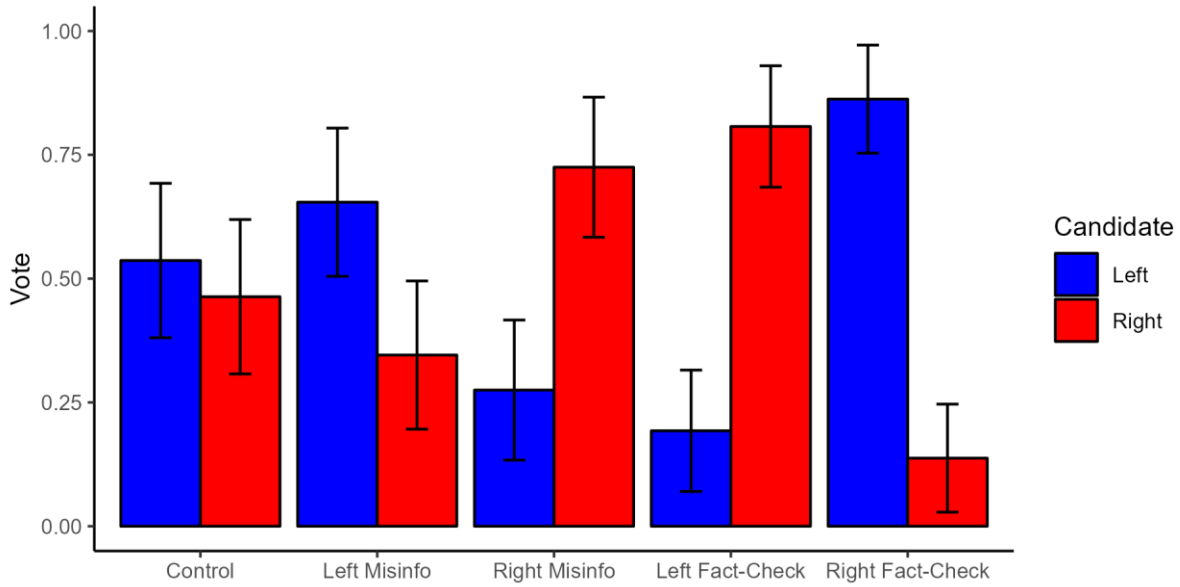
Vote

The second hypothesis stated that misinformation would increase votes for the misinformation candidate (H2A), and that exposure to fact-checks would reduce those votes (H2B), relative to control. To test this, voting outcomes were analyzed via logistic regression, with vote as the dependent variable (votes for the left-leaning candidate coded as 1, votes for the right-leaning candidate coded as 0), condition as the between-subjects predictor, and political-ideology scores as a continuous predictor. For analytical consistency, an analysis of deviance was conducted using the package *car* and planned contrasts were conducted using *emmeans* (Fox & Weisberg, 2019; Lenth, 2023). There was a main effect of condition,

$\chi^2(4) = 105.60$, $p < .001$. Figure 5 presents the mean vote likelihoods for the left- and right-leaning candidate across each condition, with voting likelihoods expressed on a 0-1 scale.

Figure 5

Mean Vote Likelihoods for Left- and Right-Leaning Candidates Across Conditions



Note. Left/Right Misinfo, condition with misinformation favoring the left/right-leaning candidate, respectively; Left/Right Fact-Check, condition with misinformation favoring the left/right-leaning candidate but fact-checked. Errors bars represent 95% confidence intervals.

Planned contrasts were conducted to test hypotheses 2A and 2B. Participants were 1.63 times more likely to vote for the left-leaning candidate in the misinformation-left condition than control, but this difference was not significant, $z = 1.53$, $p = .127$, OR = 1.63, 95% CI [0.73, 3.65]. Participants were 3.05 times more likely to vote for the right-leaning candidate in the misinformation-right condition than in control, and this difference was significant, $z = 3.34$, $p < .001$, OR = 3.05, 95% CI [1.33, 7.04]. This provides partial support for H2A. When misinformation was fact-checked, participants were 4.85 times more likely to vote for the non-misinforming candidate in the left fact-check condition compared to control, $z = 4.44$, $p < .001$, OR = 4.85, 95% CI [2.00, 11.79], and 5.42 times more likely in the right fact-check condition compared to control, $z = 4.30$, $p < .001$, OR = 5.42, 95% CI [2.03, 14.46]; this supports H2B.

In sum, misinformation boosted the likelihood of participants voting for a misinforming candidate but only when that candidate was right-wing. However, fact-checks significantly and substantially reduced voting likelihood for misinforming candidates.

False-Statement Beliefs

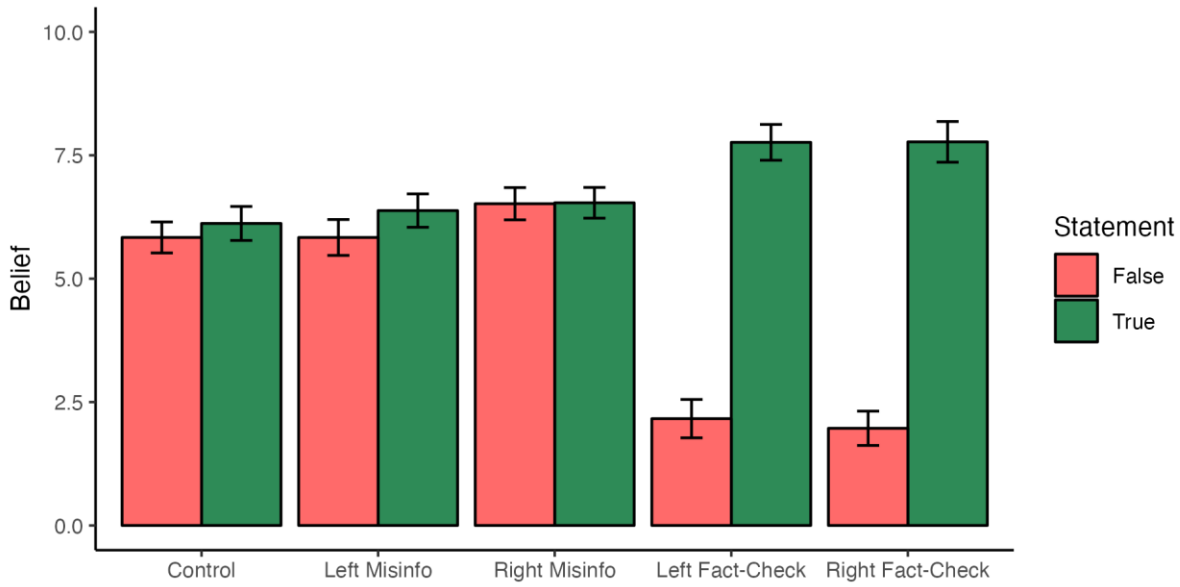
Mean belief ratings for true and false statements are presented in Figure 6. A one-way between-subjects ANOVA was conducted with condition as the factor. There was a main effect of condition, $F(4, 401) = 157.79$, $p < .001$, $\eta_p^2 = .61$, 95% CI [.56, .66].

The third hypothesis stated that belief in specific false statements would slightly increase with exposure (without being fact-checked; H3A) and that fact-checks would reduce these false beliefs (H3B), relative to control. To test this, a contrast analysis was performed. The first contrast compared control and misinformation-left conditions; this was non-significant, $t(401) = 0.002$, $p = .998$, $d < 0.01$, 95% CI [-0.31, 0.31]; thus, belief in specific false statements was not boosted by previous exposure in the misinformation-left condition compared to control. The second contrast was identical but with the misinformation-right condition; this was significant, $t(401) = 2.75$, $p = .013$, $d = 0.43$, 95% CI [0.12, 0.74]; thus, belief in specific false statements was enhanced by previous exposure in the misinformation-right condition compared to control. H3A was thus partially supported. The third contrast compared false beliefs between the control and left fact-check conditions; this was significant, $t(401) = 14.90$, $p < .001$, $d = 2.32$, 95% CI [1.97, 2.67]. The fourth contrast was identical but with the right fact-check condition; this was also significant, $t(401) = 15.55$ $p < .001$, $d = 2.44$, 95% CI [2.09, 2.80]. Thus, belief in specific false statements was significantly lower in the fact-check conditions compared to control, which supports H3B.

In sum, mere exposure to false statements slightly increased belief in statements when they favored the right-leaning candidate. Fact-checks significantly and substantially reduced false beliefs relative to control.

Figure 6

Mean Belief Ratings for True and False Statements Across Conditions



Note. Left/Right Misinfo, condition with misinformation favoring the left/right-leaning candidate, respectively; Left/Right Fact-Check, condition with misinformation favoring the left/right-leaning candidate but fact-checked. In the Control condition statements were rated but had not previously been presented. Errors bars represent 95% confidence intervals.

True-Statement Beliefs

The analogous ANOVA for true statements again yielded a main effect of condition, $F(4, 401) = 19.79$, $p < .001$, $\eta_p^2 = .16$, 95% CI [.10, .23]. The third hypothesis stated that belief in specific true statements would be increased by prior exposure (H3A) and also increased by (affirmative) fact-checks (H3C); to test this, contrast analyses were performed. The contrast analyses revealed no significant difference in true-statement belief between the control condition and either the misinformation-left, $t(401) = 1.03$, $p = .302$, $d = 0.16$, 95% CI [-0.15, 0.47], or the misinformation-right conditions, $t(401) = 1.66$, $p = .197$, $d = 0.26$, 95% CI [-0.05, 0.57]. Therefore, H3A was not supported. The third contrast compared true-statement belief between the control and left fact-check conditions. The contrast was significant, $t(401) = 6.56$, $p < 0.001$, $d = 1.02$, 95% CI [0.71, 1.34]. The fourth contrast was the same but focused on the right fact-check condition; this was also significant, $t(401) = 6.54$

$p < 0.001$, $d = 1.03$, 95% CI [0.71, 1.34]. Thus, belief in true statements was greater in the fact-check conditions than control, and H3C was supported.

In sum, mere exposure to true statements did not increase belief in them relative to control, but fact-checks slightly but significantly increased belief relative to control.

Discussion

The present study used a simulated election to investigate the impact of misinformation and fact-checks on voter perceptions, feelings, behaviors, and beliefs. There was some evidence that, relative to a control condition, misinformation favoring a candidate could sometimes be beneficial, primarily for right-leaning candidates. Specifically, when there was misinformation favoring the right-leaning candidate and no fact-checking occurred, the right-leaning candidate was perceived as more competent and more moral, participants felt more warmly toward them, and participants were more likely to vote for them relative to the left-leaning candidate. In contrast, when there was misinformation favoring a left-leaning candidate, this only led to them being perceived as significantly more moral than their political opponent. Additionally, although there were some minor benefits of misinformation, the effects of fact-checking were much larger. When statements were fact-checked, participants viewed candidates who spread misinformation much less favorably than their political opponents, perceiving them as less competent and less moral, and having reduced feelings of warmth toward them. Crucially, this also translated to behavior, with fact-checking leading to a decreased probability of voting for candidates who spread misinformation. This suggests that although politicians may be tempted to spread misinformation that paints them in a positive light or denigrates their opponent, there is considerable potential downside if they are fact-checked.

Impacts of Misinformation and Fact-Checks

Misinformation that was not fact-checked had greater influence on feelings, perceptions, voting behavior, and beliefs when it favored right-leaning as opposed to left-leaning candidates. One potential explanation for this asymmetry may relate to participants' belief in the false statements, which was greater than control only if the false statements favored the right-leaning candidate. Although the statements deemed to be misleading were designed to be politically moderate, they aimed to increase relative perceptions of competence, which may be perceived as being boastful or denigrating of their political opponent; some misleading statements also made direct appeals to the 'people'. These are characteristics that are typical of statements made by 'populist' leaders (see Bracciale & Martella, 2017; Rice-Oxley & Kalia, 2018). Therefore, given the rising prominence of right-wing populism globally (Rodrik, 2021; Rosenberg, 2023), it is possible that the misleading statements seemed more plausible when coming from a right-leaning politician, particularly to U.S.-based participants in the (post-) Trump era (Nai et al., 2019; Oliver & Rahn, 2016; Schneiker, 2020). There is also some evidence that right-leaning voters may be more likely to be impacted by misinformation, and that right-wing politicians are more likely to spread misinformation (Baptista & Gradim, 2022; Bode et al., 2018; Frischlich et al., 2021; Hameleers & Minihold, 2020; van Kessel et al., 2020). Cumulatively, our results suggest that misleading statements may have greater utility for politicians on the political right, but further research is needed to examine whether different forms or styles of misinformation differentially advantage politicians across the political spectrum.

Unsurprisingly, and consistent with previous research, fact-checking led to reduced belief in false statements and increased belief in true statements (Aird et al., 2018; Berinsky, 2017; Prike et al., 2023; Swire, Berinsky, et al., 2017; Swire-Thompson et al., 2020). There were large and consistent effects of fact-checking, with reduced feelings of warmth,

perceptions of competence and morality, and voting for both left- and right-leaning politicians who spread misinformation that was fact-checked. The effects found were much larger than those obtained in previous studies that used well-known real-world politicians (Aird et al., 2018; Leyva & Beckett, 2020; Prike et al., 2023; Swire, Berinsky, et al., 2017; Swire-Thompson et al., 2020). This supports the notion that one reason prior studies have found little or no attitudinal change is due to strong pre-existing attitudes. However, the results suggest that in the absence of strong prior attitudes, voters view candidates who spread misinformation negatively, and are willing to punish them at the ballot box.

There are several important practical implications of our findings regarding the efficacy of fact-checking. First, it suggests that political newcomers, about whom voters have not yet developed strong attitudes, are the ones most likely to suffer negative consequences if they are fact-checked and caught spreading misinformation. A related implication is that fact-checking politicians' statements prior to strong attitudes about them being formed will have greater impact on tangible support measures and may thus serve to improve the political information environment. New voters (e.g., those reaching voting age, moving to a new country or electorate) and political moderates in particular may be receptive to updating their attitudes in response to fact-checks and may be more easily dissuaded from voting for politicians who spread misinformation, given they may hold weaker pre-existing attitudes. In a relevant field study, Nyhan and Reifler (2015) found that U.S. state representatives made fewer false claims after they received a letter reminding them of the potential reputational consequences of receiving a negative fact-check rating. Therefore, communicating these findings to politicians and highlighting that the potential downsides of being caught spreading misinformation may be greater than previously thought, could help to discourage politicians from spreading misinformation.

Limitations and Future Directions

Although this study is the first to examine how moderate voters (a highly influential segment of U.S. voters; Fowler et al., 2023; Saad, 2022) update their impressions of political candidates in response to misinformation and fact-checks, there are also some limitations and areas for future work. A key limitation is that the present study was a single-session experimental design, with minimal delay between exposure to misinformation, fact-checks, and measures. However, in the real world voters typically encounter pieces of misinformation and/or fact-checks irregularly over extended periods of time, and then later make voting decisions in an environment where they will be influenced by many other factors. The impact of misinformation and/or fact-checks on real-world voting behavior will therefore be smaller, particularly if voters are not repeatedly exposed (Swire, Ecker, et al., 2017; Swire-Thompson et al., 2023). Future studies may enhance ecological validity by conducting a simulated election over an extended period, with participants exposed to misinformation and/or fact-checks across multiple sessions, prior to voting.

Incorporating fictional political candidates enabled our study to determine the effects of misinformation and fact-checking in a situation devoid of pre-existing attitudes. However, the fictional nature of the materials naturally limits the generalizability of the results to real-world elections. Future studies looking to examine misinformation and fact-checking effects may benefit from either using real-world politicians who voters are unfamiliar with, or by incorporating real-world politicians who voters are familiar with, but undecided on. Additionally, we intentionally included politically moderate participants, and the fictional candidates and political messages included within the study were only slightly left or right-leaning. Therefore, it would also be beneficial for future research to include participants, political candidates, and/or messages that cover a broader range of political views to examine whether and at what point ideological alignment (or opposition) reduces the impact of

misinformation and fact-checking. Such studies would provide a valuable middle ground between the present study and prior studies focused on real, well-known politicians and partisan participants.

Conclusion

Overall, our findings suggest that politicians who voters are not familiar with may slightly benefit from misinformation that presents them in a positive light (relative to an opponent), particularly if they are right-leaning. However, these minor benefits are vastly outweighed by the potential reputational risk associated with being fact-checked.

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Declarations

Data Availability: This study was not pre-registered. Data, analysis scripts, and materials from the empirical studies are available on the Open Science Framework at https://osf.io/wfbck/?view_only=959ded2da87441d19f7498a20dfebc04.

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